

Volume I Foundational Geometry Reference

Contents

0.1	Foundations	4
0.2	Compass-and-Straightedge Constructions	4
0.3	Analytical Geometry	6
0.4	Lines	6

Euclidean Geometry

Definitions

Definition 0.0.1 (Point). A point is that which has no part.

Definition 0.0.2 (Line). A line is breadthless length.

Definition 0.0.3 (Ends of a line). The ends of a line are points.

Definition 0.0.4 (Straight line). A straight line is a line which lies evenly with the points on itself.

Definition 0.0.5 (Surface). A surface is that which has length and breadth only.

Definition 0.0.6 (Edges of a surface). The edges of a surface are lines.

Definition 0.0.7 (Plane surface). A plane surface is a surface which lies evenly with the straight lines on itself.

Definition 0.0.8 (Plane angle). A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

Definition 0.0.9 (Rectilinear angle). And when the lines containing the angle are straight, the angle is called rectilinear.

Definition 0.0.10 (Right angle and perpendicular). When a straight line standing on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.

Definition 0.0.11 (Obtuse angle). An obtuse angle is an angle greater than a right angle.

Definition 0.0.12 (Acute angle). An acute angle is an angle less than a right angle.

Definition 0.0.13 (Boundary). A boundary is that which is an extremity of anything.

Definition 0.0.14 (Figure). A figure is that which is contained by any boundary or boundaries.

Definition 0.0.15 (Circle). A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure equal one another.

Definition 0.0.16 (Center of a circle). And the point is called the center of the circle.

Definition 0.0.17 (Diameter). A diameter of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

Definition 0.0.18 (Semicircle). A semicircle is the figure contained by the diameter and the circumference cut off by it. And the center of the semicircle is the same as that of the circle.

Definition 0.0.19 (Rectilinear figures). Rectilinear figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.

Definition 0.0.20 (Equilateral, isosceles, and scalene triangles). Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.

Definition 0.0.21 (Right-angled, obtuse-angled, and acute-angled triangles). Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acute-angled triangle that which has its three angles acute.

Definition 0.0.22 (Quadrilateral classes). Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

Definition 0.0.23 (Parallel straight lines). Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

Postulates

0.1 Foundations

Axioms

Propositions

Theorem 0.1.1 (Euclid I.1: Constructing an equilateral triangle). *Given a finite straight line segment AB , there exists an equilateral triangle ABC constructed on AB .*

Theorem 0.1.2 (Euclid I.2: Copying a segment from a given point). *Given a point A and a finite straight line segment BC , there exists a straight line segment beginning at A and equal to BC .*

Theorem 0.1.3 (Euclid I.3: Cutting off an equal segment). *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

0.2 Compass-and-Straightedge Constructions

Proofs

Trigonometry

Proofs

Analytical Geometry

0.3 Analytical Geometry

0.4 Lines

Definition 0.4.1 (Point-slope form). Let $P_1 = (x_1, y_1)$ be a point in the Euclidean plane and let $m \in \mathbb{R}$. The point-slope form of the line through P_1 with slope m is

$$y - y_1 = m(x - x_1).$$

Definition 0.4.2 (Two-point form). Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be distinct points with $x_1 \neq x_2$. The two-point form of the line through P_1 and P_2 is

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1).$$

Definition 0.4.3 (General linear form). The general linear form of a line in $\mathbb{R}^2$ is

$$Ax + By + C = 0,$$

where $A, B, C \in \mathbb{R}$ and $A^2 + B^2 \neq 0$.

Definition 0.4.4 (Vector parametric form). A line in $\mathbb{R}^2$ may be represented as the image of $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$ given by

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v},$$

where $\mathbf{r}_0 \in \mathbb{R}^2$ is a point on the line and $\mathbf{v} \in \mathbb{R}^2$ is a nonzero direction vector.

Definition 0.4.5 (Intercept form). If a line meets the x -axis at $(a, 0)$ and the y -axis at $(0, b)$ with $a, b \neq 0$, its intercept form is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

Definition 0.4.6 (Affine parametric line in $\mathbb{R}^n$). Given distinct points $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the line through them is

$$\mathbf{x}(t) = \mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1), \quad t \in \mathbb{R}.$$

Definition 0.4.7 (Normal form). A line in $\mathbb{R}^2$ may be written in normal form as

$$x \cos \alpha + y \sin \alpha - p = 0,$$

where $p \geq 0$ is the distance from the origin to the line.

Definition 0.4.8 (Symmetric form). Given a point (x_1, y_1) and direction cosines $(\cos \theta, \sin \theta)$, the symmetric form of a line is

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r.$$

Proofs